Maxwell Tsai

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EDUCATION

Boston University – College of Engineering Expected May 2027
Bachelor of Science in Electrical Engineering, Concentration in Robotics GPA: 3.96/4.00

Coursework: Robotics, Control Systems, Computational Optical Imaging, Digital Image Processing (Spring 2027), Engineering Mechanics, Signals and Systems, Probability and Statistics, Software Engineering

PROJECTS

BusBot: Autonomous Dish-Collection Robot May 2026 – Present

- Led a 6-person team (2 EE, 2 CE, 2 ME) for the first 2 months, defining the system architecture (ESP32-S3 sensor controller, Jetson Orin Nano running ROS 2 for LiDAR navigation) and requirements from a 25-diner survey
- Defined sensor coverage for LiDAR navigation, adding ultrasonic sensing to cover the 2D 276º LiDAR's rear blind spot within a 0.2 m obstacle buffer
- Wrote ESP32-S3 firmware (ESP-IDF, C) for obstacle stopping with PWM motor control and quadrature encoder feedback: brakes at 40 cm, resumes past 45 cm, and stops on sensor failure
- Validated on a motor bench rig: ±2 cm range accuracy, brake command under 1 ms after detection, 5/5 stop/restart trials
- Sized a 12 V, 18 Ah SLA battery for the 2.5 hr runtime requirement from a component-level power budget (~35 W draw)

Autonomous Laundry Folder Jan 2026 – May 2026

- Integrated power and control for a SCARA-inspired 2R1P manipulator (laser-cut parallel-plank linkage, aluminum extrusion frame) as the sole EE on a 5-person team
- Derived forward kinematics by hand and co-planned a 10-step folding trajectory across a 70×60×20 cm workspace
- Debugged Arduino joint control against a teammate's inverse kinematics solution, reaching correct joint angles at each step
- Powered two 25 kg·cm linkage servos and an electromagnet end effector from a 12 V supply through buck converters

Bedside Beacon: Point-to-Point Communication Device Sep 2025 – Dec 2025

- Built a servo-actuated laser transmitter with a 90° steerable beam and repeatable return-to-center positioning
- Tested distances from 4 to 15 ft with 7 trials each; 29 of 35 transmissions (83%) succeeded beyond the 5 ft design spec

EXPERIENCE

Intel Hillsboro, OR
Data Science and Analytics Intern May 2026 – Aug 2026

- Cut wafer data retrieval from 2+ weeks to 2 hours (~100 measurements, ~200 wafers) by building a data-pulling tool in JMP Scripting Language with device and automation teams; adopted by the team after the internship
- Traced errors in a ridge regression model predicting in-line electrical test values to outdated metrology revisions in the training data; the tool keeps only the latest revision per measurement per wafer

Analog Product Development & Test Engineer Intern May 2025 – Aug 2025

- Architected an object-oriented Python API abstracting 4 lab instruments, including a Bit Error Rate Tester, as generalized scopes, clocks, and sources, letting engineers test remotely with common functions instead of instrument-specific commands
- Automated IEEE 488.2 SCPI communication over GPIB (PyVISA) to configure instruments and transfer measurement data
- Refactored 7,000+ lines of Python to PEP 8 standards and wrote ~100 unit tests covering API functions for each instrument

Boston University Dept. of Electrical and Computer Engineering Boston, MA
Teaching Assistant – Logic Design Jan 2026 – May 2026

- Led Verilog FPGA lab sections in Vivado and debugged student designs; graded homework and exams for 80+ students

SemeterLab at Boston University Boston, MA
Undergraduate Research Assistant Sep 2025 – Dec 2025

- Developed an Android app logging Global Navigation Satellite System (GNSS) measurements for real-time ionospheric monitoring, tracking 15+ satellites and exporting carrier-to-noise density (C/N0) data

SKILLS

Robotics: kinematics, trajectory planning, motor control (PWM, quadrature encoders), sensor integration (ultrasonic, v/i sensing)
Embedded: ESP32-S3 (ESP-IDF), Arduino, I2C
Programming: Python, C, C++, MATLAB
Fabrication: KiCAD, Onshape (CAD), laser cutting, 3D printing

LEADERSHIP

Theta Tau (New Member Education, Professional Development and Brotherhood Chair): ran resume and interview workshops and coached 20+ members 1-on-1 | **BostonU Club Tennis** (Community Service Chair): organized youth field day service event